

Keast Group Protocols

Placental alkaline phosphatase (PLAP) staining of bladder sections

Preparation of tissue for cryosectioning

1. Dissect bladder out fresh from the mouse.
- NOTE: Perfusion of mouse can lead to loss of alkaline phosphatase staining
2. Tissue fixation: Embryonic tissue - fix tissue in 4% PFA overnight at 4°C.
Postnatal tissue - fix tissue in 4% PFA for 3-4hrs at 4°C.
3. Wash 3 x 30 mins in 0.1M PBS (pH 7.2) at room temperature.
4. Wash and store tissue in 0.1M PBS (pH 7.2) at 4°C (containing 0.1% azide if storing for longer than 1 week).
5. The day before cutting place tissue in 30% sucrose in PBS (0.1M, pH 7.2), leave overnight at 4°C.
6. Embed tissue in OCT and freeze. Cut sections on cryostat, mounting sections on 1% gelatinised slides.

Staining of tissue

1. Air dry slides at room temperature for 30 mins.
2. Wash slides 3 x 10 mins in 0.1M PBS (pH 7.2) at room temperature.
- NOTE: For all ages besides adult go straight to step 5. For adult bladder sections perform Steps 3 and 4 to abolish endogenous alkaline phosphatase staining of the urothelium. There may be a small loss of nerve staining in the bladder after treatment with acetic acid.
3. For adult bladder only: Dip slides in 20% acetic acid in dH₂O at 4°C for 10 secs.
4. Wash slides 3 x 10 mins in 0.1M PBS (pH 7.2) at room temperature.
5. Heat inactivate endogenous alkaline phosphatase by incubating the slides in pre-heated 0.1M PBS (pH 7.2) at 72°C for 90 mins.
6. Rinse slides in Alkaline Phosphatase Buffer 1 x 5 mins, then 1 x 10 mins at room temperature.
7. Freshly prepare the Staining Solution by adding 20µl of NBT/BCIP Stock Solution (Roche, Cat# 1681451) per 1 ml of Alkaline Phosphatase Buffer. Incubate slides in Staining Solution overnight in the dark at room temperature.
8. Rinse slides 3 x 10 mins in 0.1M PBS (pH 7.2) at room temperature.

9. Coverslip slides in buffered glycerol. Seal edges of coverslip with nail polish.

Solutions

Solutions A and B (below) are used for the making of PBS:

Solution A: (0.2M): 24.0g NaH_2PO_4 / 1000 ml H_2O

Solution B: (0.2M): 28.4g Na_2HPO_4 / 1000 ml H_2O

Phosphate Buffered Saline (PBS) 0.1M (pH 7.2)

1. 140 ml Sol A + 360 ml Sol B + ~450 ml H_2O + 8.5g NaCl. Dissolve.
2. pH to 7.2 with 1M NaOH/HCl
3. Make up to 1 litre with H_2O

Alkaline Phosphatase Buffer (pH 9.5)

12.11g Tris Base (100mM)

5.84g NaCl (100mM)

10.17g $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (50mM)

Make up to 1 litre with H_2O

pH to 9.5 with 1M HCl